

Westinghouse – AP1000[®] Plant Overview

Latin American Section of the American Nuclear Society
Brazil – July 2014

Ed Cummins

Vice President & Chief Technologist, New Plant Technologies

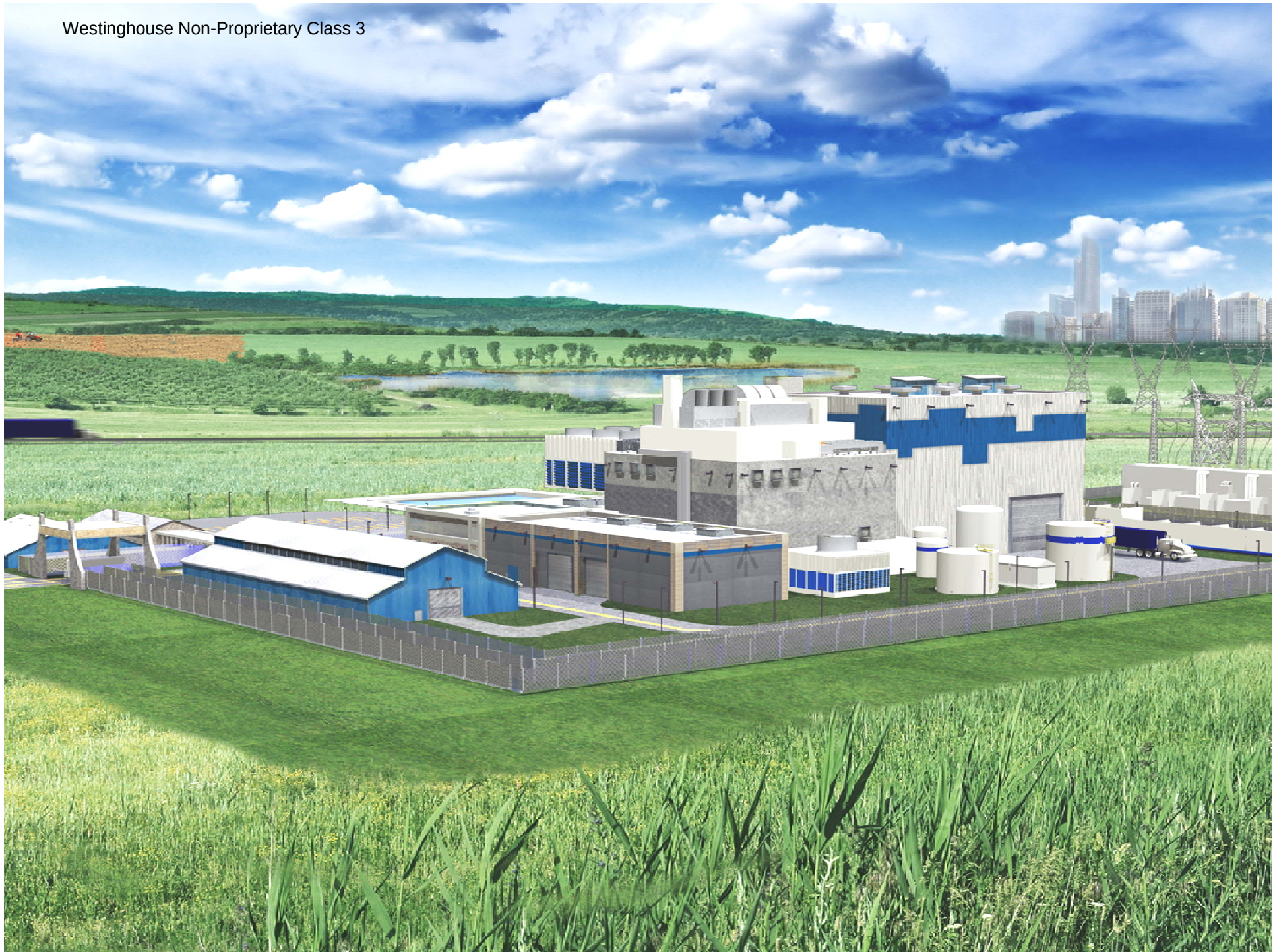
AP1000 is a trademark or registered trademark of Westinghouse Electric Company LLC, its affiliates and/or its subsidiaries in the United States of America and may be registered in other countries throughout the world. All rights reserved. Unauthorized use is strictly prohibited. Other names may be trademarks of their respective owners.



Westinghouse SMR overview

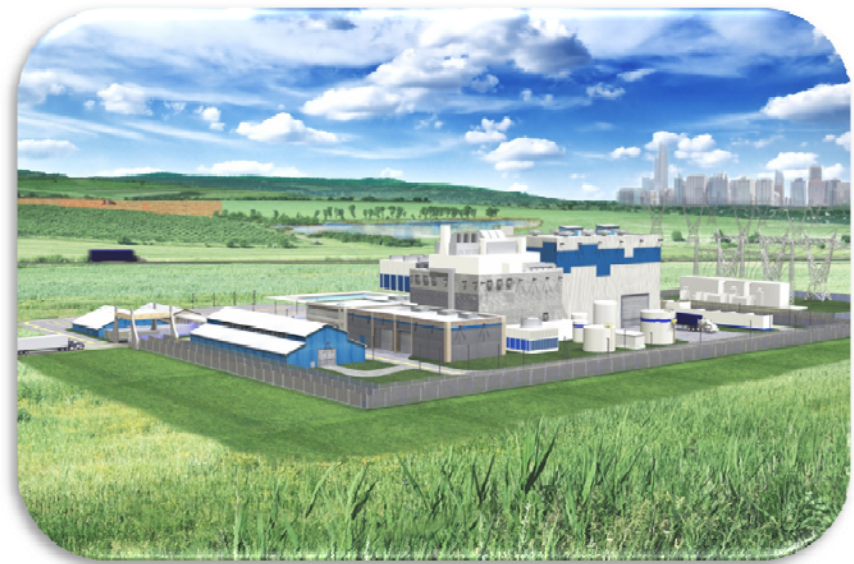


Westinghouse Non-Proprietary Class 3

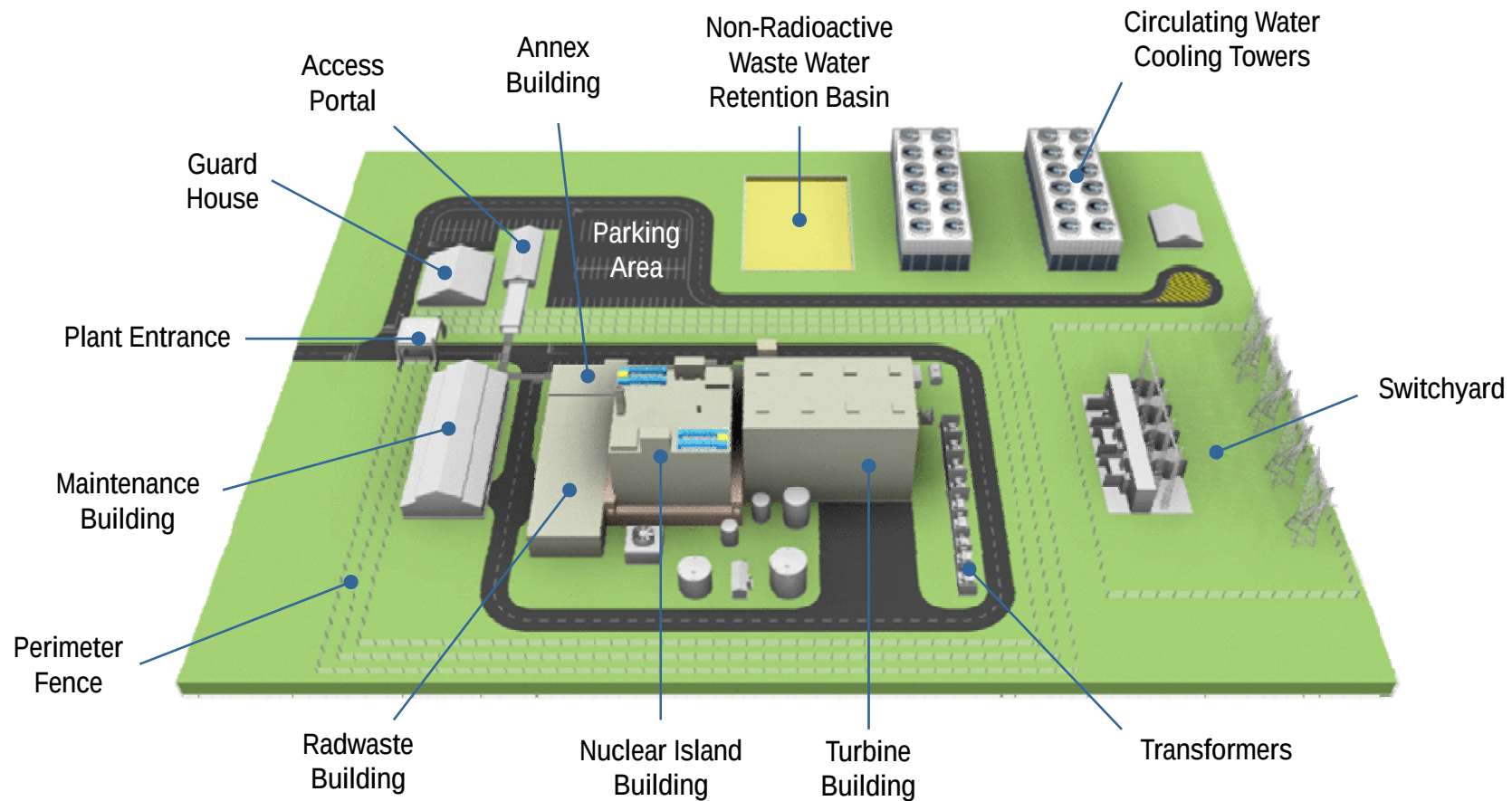


Westinghouse SMR Plant Features

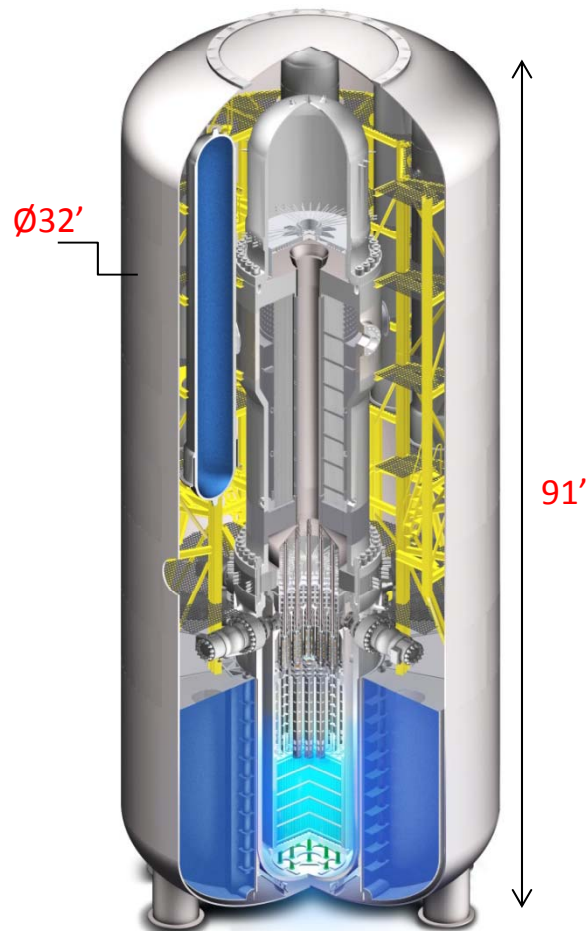
- **Integral Pressurized Water Reactor (PWR) with innovative packaging of industry-proven components**
- **Compact 15 acre plant site, with containment and reactor vessels below grade**
- **Highest levels of safety, with 100% reliance on natural forces for indefinite passive core cooling**
- **Reduced licensing risk assured through AP1000[®] plant experience**
- **100% modular construction with 8-month NOAK construction schedule**



Westinghouse SMR Site Plan



Westinghouse SMR Technology Features



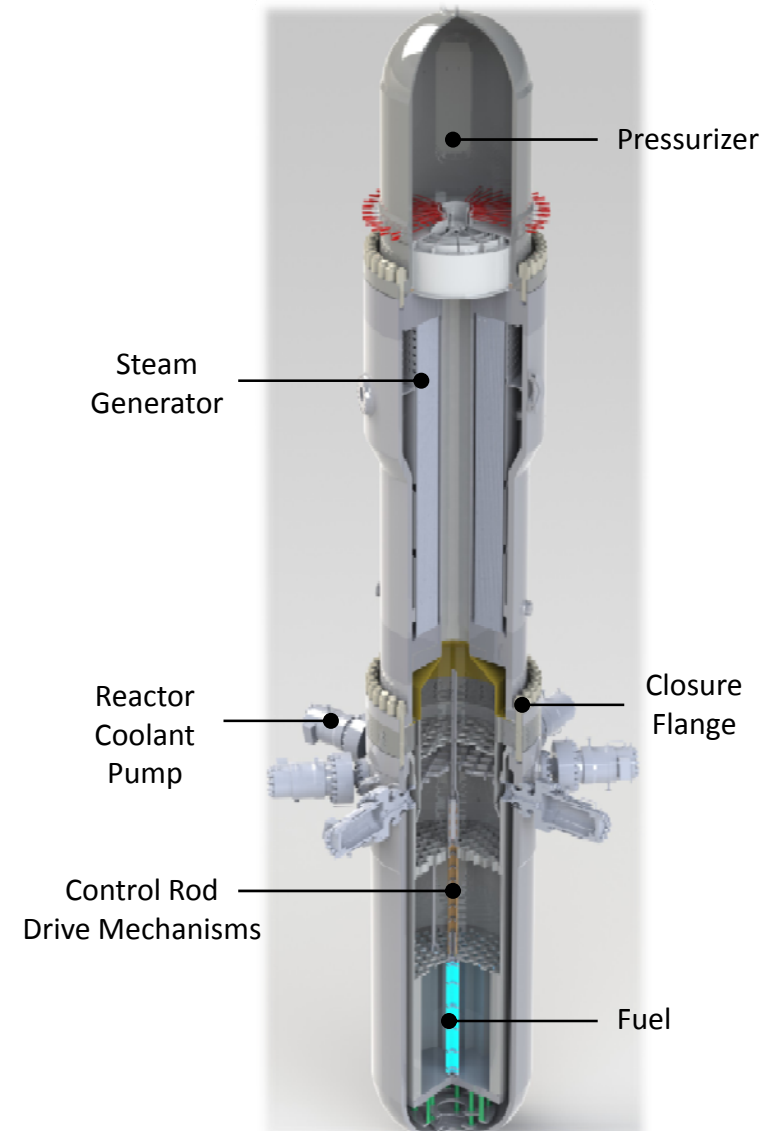
Specifications

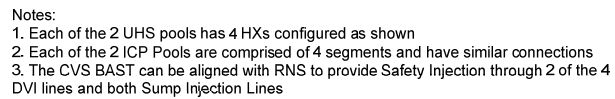
Thermal Output	800 MWt
Electrical Output	>225 Mwe
Passive Safety Systems	No operator intervention required for 7 days
Core Design	17x17 Robust Fuel Assembly 8.0 ft. Active Length < 5% Enriched U235 89 Assemblies Soluble Boron and 37 Internal CRDMs 24-Month Refueling Interval
Reactor Vessel Size	Outer Diameter: 12 ft. Height: 81 ft.
Upper Vessel Package	280 Tons
Containment Vessel Size	Outer Diameter: 32 ft. Height: 91 ft. Fully Modular Construction
Reactor Coolant Pumps	8 External, Horizontally-Mounted Pumps Sealless Configuration
Steam Generator	Recirculating, Once-Through, Straight-Tube
Pressurizer	Integral to Vessel
Instrumentation and Control	Ovation®-based Digital Control System

Reactor Design

- ASME Section III Subsection NB Vessel
- 12 ft max flange diameter (13 ft shipping)
- Reactor height ~85 ft
- Design flow rate of 100,000 gpm
- 37 internal control rods (shutdown/control)
- Integral reactor; no large-bore loop piping
- Straight tube Steam Generator (SG) with external circulation
- No penetrations below top of core
 - In-vessel retention for severe accidents

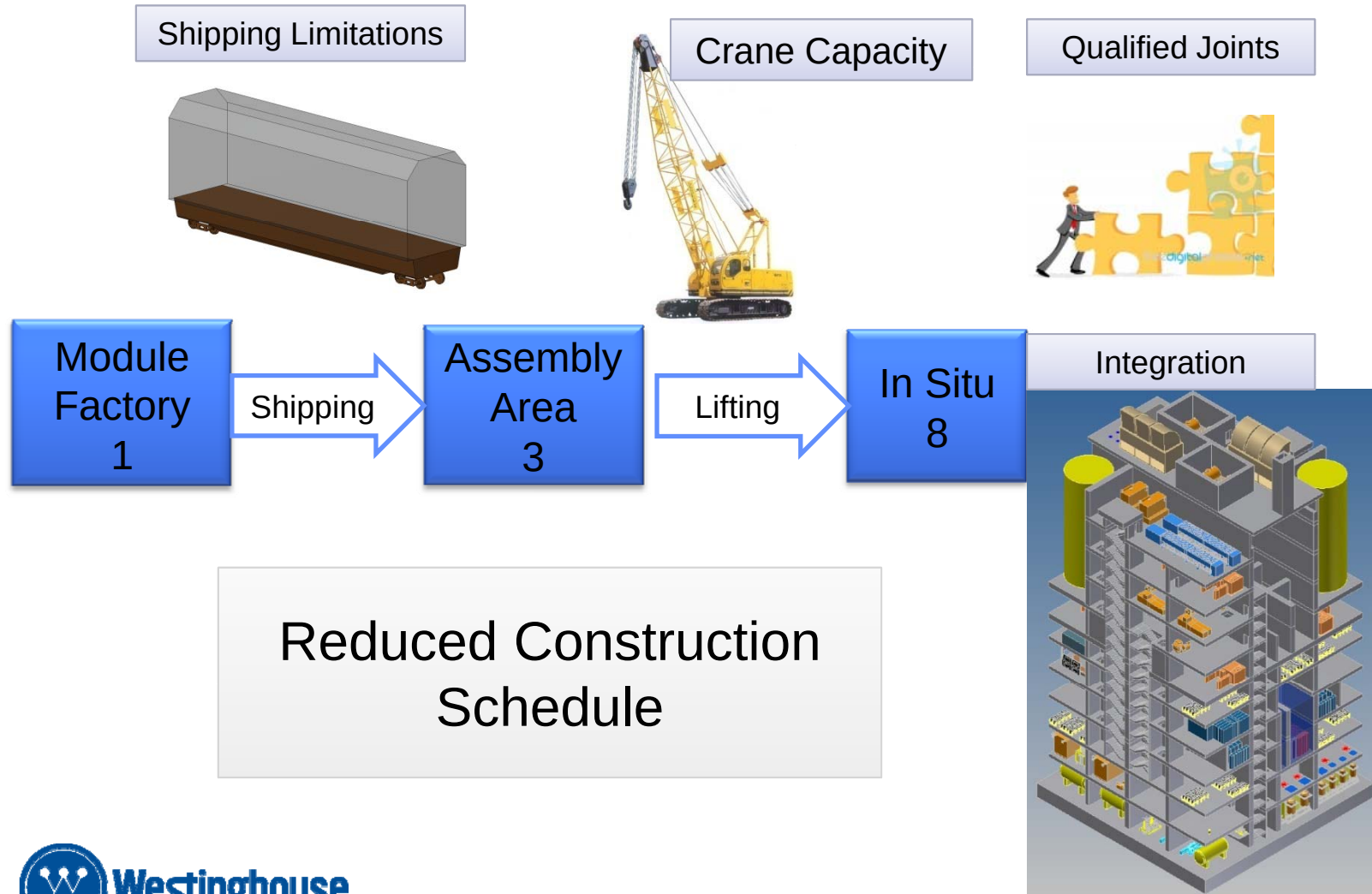
**Proven, Familiar Standard
PWR Technology**





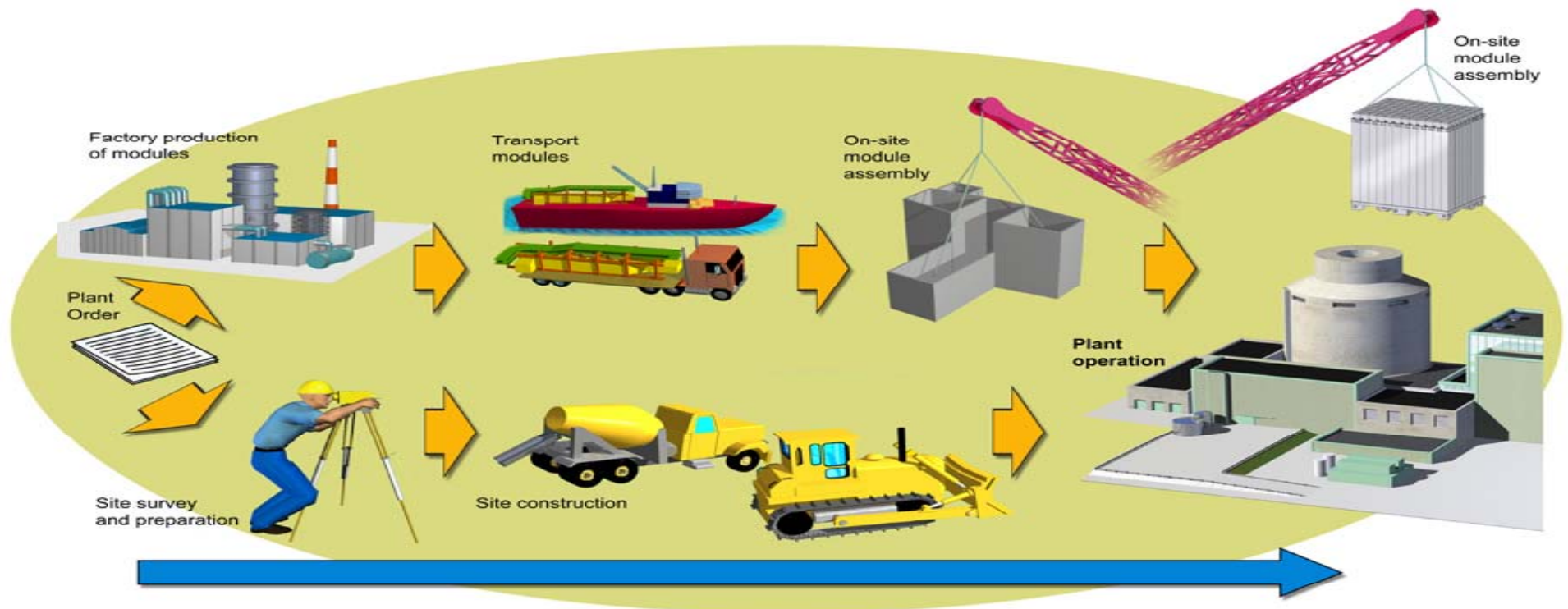
Why Modules

Modular Design Goal: Increase Factory Fabrication and Reduce Schedule



AP1000 Plant Greater Project Certainty and Shorter Schedule

Modular Construction means more work done in parallel.



Leads to shorter schedule and improved quality.

AP1000 Plant Modules

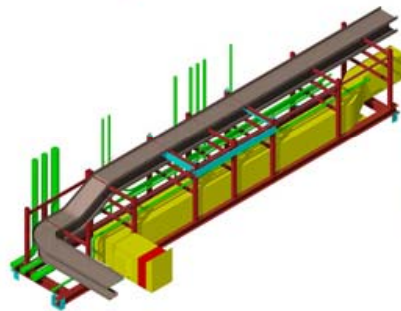




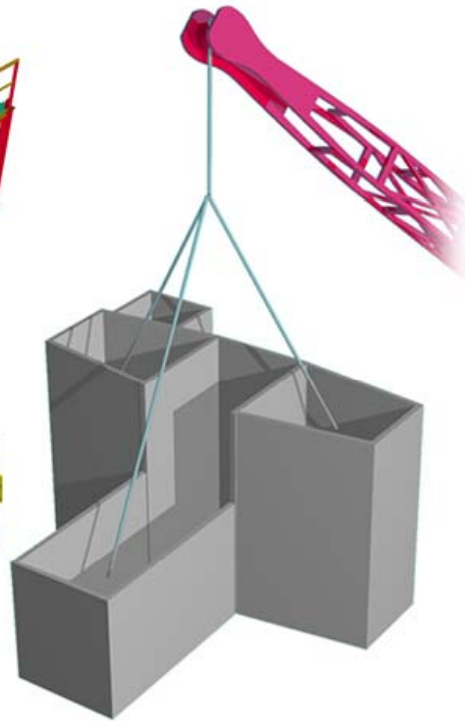
AP1000 Plant

Modules Designed into AP1000 Plant from the Beginning

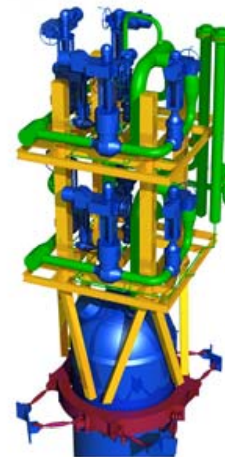
Pump/Valve Module



Raceway Module



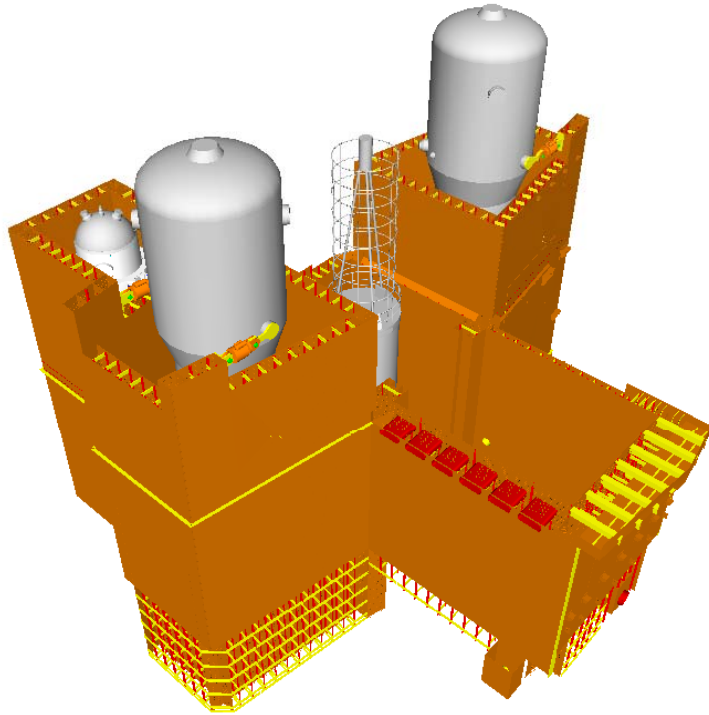
Structural Module



Depressurization Module

<u>Module Type</u>	<u>Number</u>
Structural	122
Piping	154
Mechanical Equipment	55
Electrical Equipment	11
TOTAL	342

CA01 Steam Generator & Refueling Canal Inside Containment Composite Wall Module



Steam Generators, Pressurizer
and Integrated Head Package
shown inside CA01



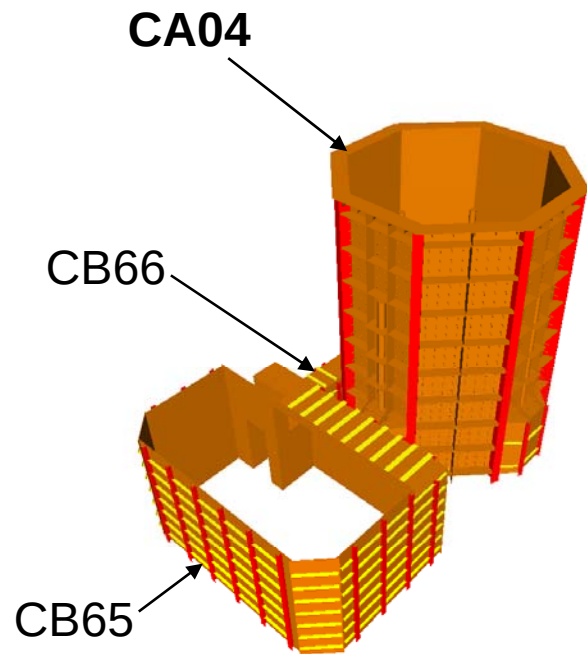
CA01 Comprised of
47 Sub-Modules:

Size (L x W x Height):
92'-0" x 96'-0" x 76'-0"
[28 m x 29.3 m x 23.2 m]

Lift Weight:
2,357,000 lbs. [1,069 MT]



CA04 Reactor Vessel Cavity/Reactor Coolant Drain Tank Inside Containment Composite Wall Module



CB65 & CB 66
Concrete Form Modules

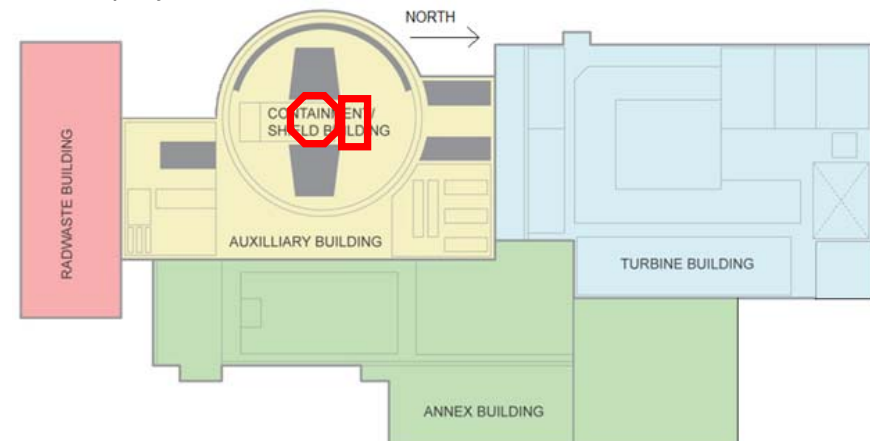


©2013 Sanmen Nuclear Power Company Ltd. All rights reserved.

CA04 Comprised of
5 Sub-Modules:

Size (L x W x Height):
21'-1" x 21'-1" x 26-6"
[6.4m x 6.4 m x 8.1m]

Lift Weight:
79,200 lbs. [35.9 MT]



CA20 Auxiliary Bldg Area 5 and 6 Module Auxiliary Building Composite Wall Module



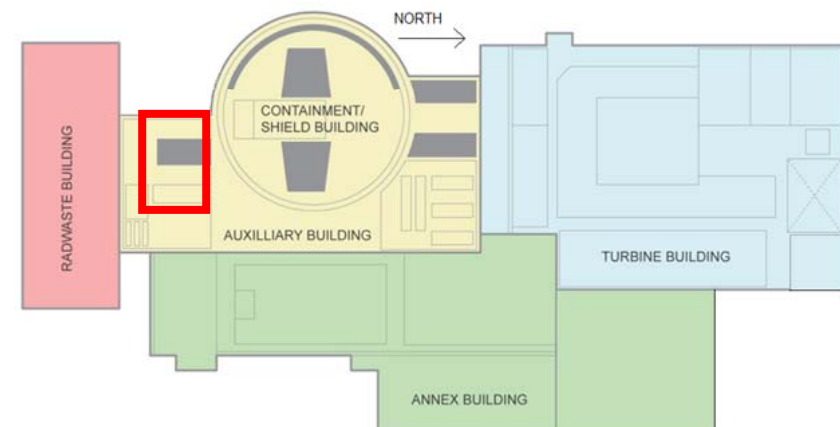
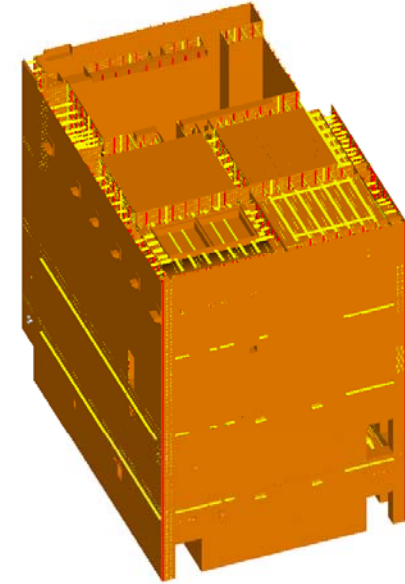
©2013 Georgia Power Company. All rights reserved.

CA20 Comprised of 72 Sub-Modules:

Size (L x W x Height):
67'-3" x 46'-6" x 68'-9"

[20.5m x 14.2m x x 21m]

Weight :(Equipment &
Mechanical Module outfitting
is not included)
1,996,000 lbs. [905 MT]



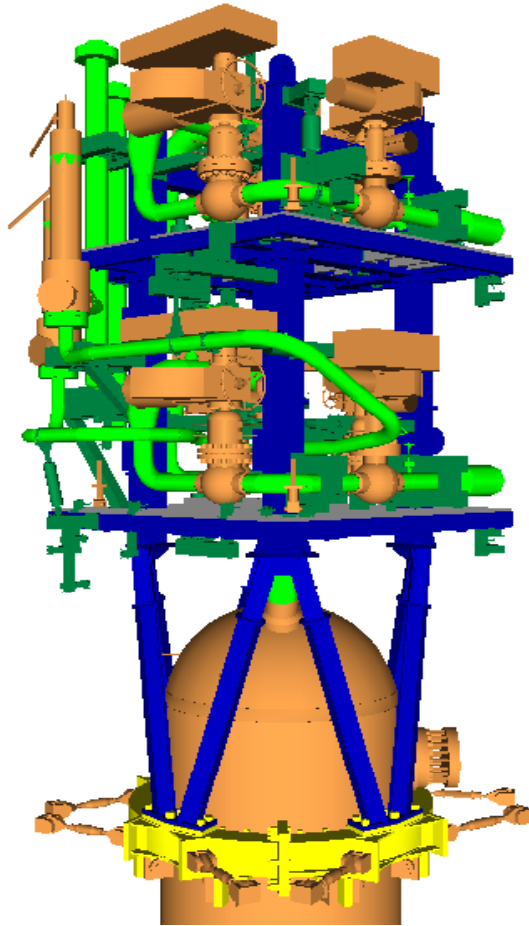
CA20 at VC Summer

CVBH at VC Summer

CV Ring at VC Summer

AP1000 Plant Major Mechanical Modules

Q601 RCS Module, ADS Stages 1, 2, 3



Size (L x W x Height):
16'-10" x 14'-4" x 32'-1"

[5.13m] x [4.37m] x [9.78m]

Lift Weight:
W/out Transport Skid

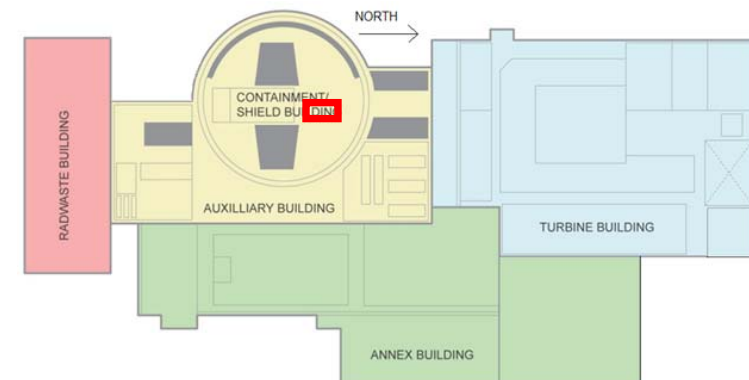
94,400 lbs. [42,819 kg]

W/ Transport Skid

102,120 lbs. [46,320]

Classification:
Safety Class A

Seismic Cat-I



AP1000 Plant Integrated Head Package (IHP) Middle and Upper Shroud



©2013 Sanmen Nuclear Power Company Ltd. All rights reserved.



©2013 Sanmen Nuclear Power Company Ltd. All rights reserved.

AP1000 Plant Lessons Learned

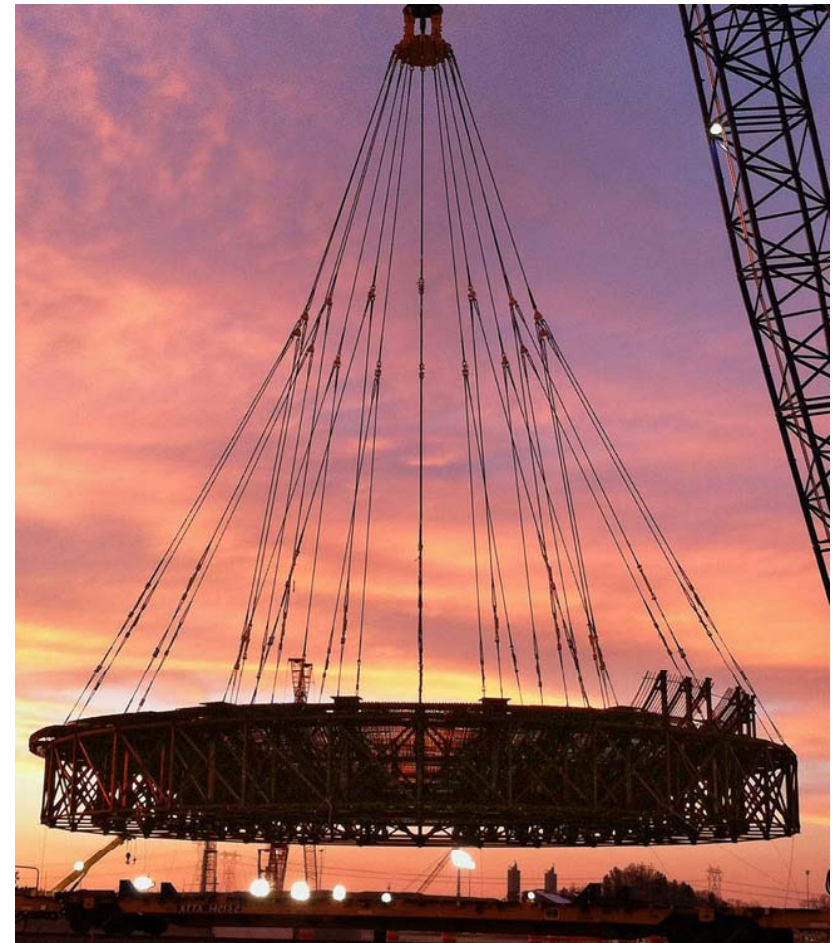


Lessons Learned Presentation Objectives

- Westinghouse Continuous Learning Process
- Lessons Learned Overview
- Examples of Lessons Learned



©2013 Georgia Power Company. All rights reserved.



©2013 Sanmen Nuclear Power Company Ltd. All rights reserved.

AP1000 Plant Module Lessons Learned (Fabrication)

- Challenge:
 - First-of-a-Kind Fabrication, experienced significant learning curves in fabrication of structural and mechanical modules
- Lessons Learned:
 - Fabricator(s)
 - Pursue Fabricators with Strong Nuclear Culture and Experience with Module Building Processes and Techniques
 - Module Team to be Engaged with Fabricator Early and Often
 - Fabricator to Produce Mockups to Demonstrate Compliance and Improve Processes
 - Westinghouse Personnel Gaining Valuable Experience Working with Fabricators on Current China and U.S. Projects
 - Stud Welding – China Initially used Low Priced Machines with Marginal Controls and resulted in Better Machines with Better Controls
 - Design Tolerances – Validated in China



AP1000 Plant Module Lessons Learned (Site Assembly)

- Challenge:
 - Tolerance “stack-up” and Quality Control have presented challenges during assembly and installation of structural modules
- Lessons Learned:
 - Fabricator to be Responsible for Site Assembly
 - Fabricator to Assemble Mockups to Demonstrate Compliance and Improve Processes
 - Westinghouse Personnel Gaining Valuable Experience Working with Fabricators on Current China and U.S. Projects
- Climate Protection for Site Assembly
 - Module Assembly Building
 - Building Size & Crane Capacity
- Assembly Methods
 - China – Manually Oriented
 - Sub Modules (Horizontal)
 - Sub Assemblies (Vertical)
 - U.S. – Automation Oriented
 - Sub Modules (Vertical)
 - Sub Assemblies (Vertical)



©2013 Georgia Power Company. All rights reserved.

AP1000 Plant Module Lessons Learned (Installation)

- Design Modifications
 - Dowels Interfered with Module During Install
 - China – Solution was Proper Alignment of Dowels
 - U.S. – Base Concrete Connection Redesigned for Ease of Installation



©2013 Sanmen Nuclear Power Company Ltd. All rights reserved.

AP1000 Plant Containment Vessel Lessons Learned

- Mature Integrated Project Schedule (IPS)
 - Developed for China
 - Further Refined for U.S. Projects
 - CV Bottom Head Installed 2 Months After FNC
- Site Interfaces
- Integration with Engineering, Quality, Supply Chain, Constructor, Customer, Etc.



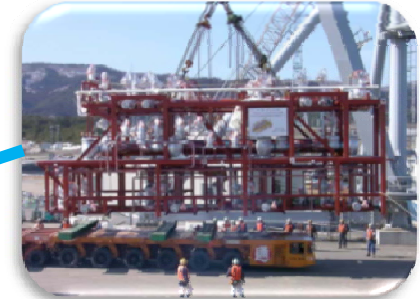
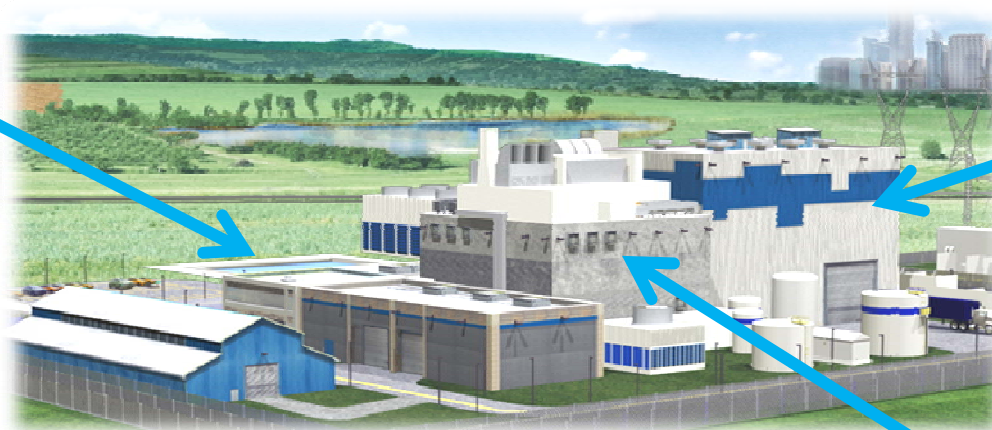
©2013 Georgia Power Company. All rights reserved.

SMR Modularization

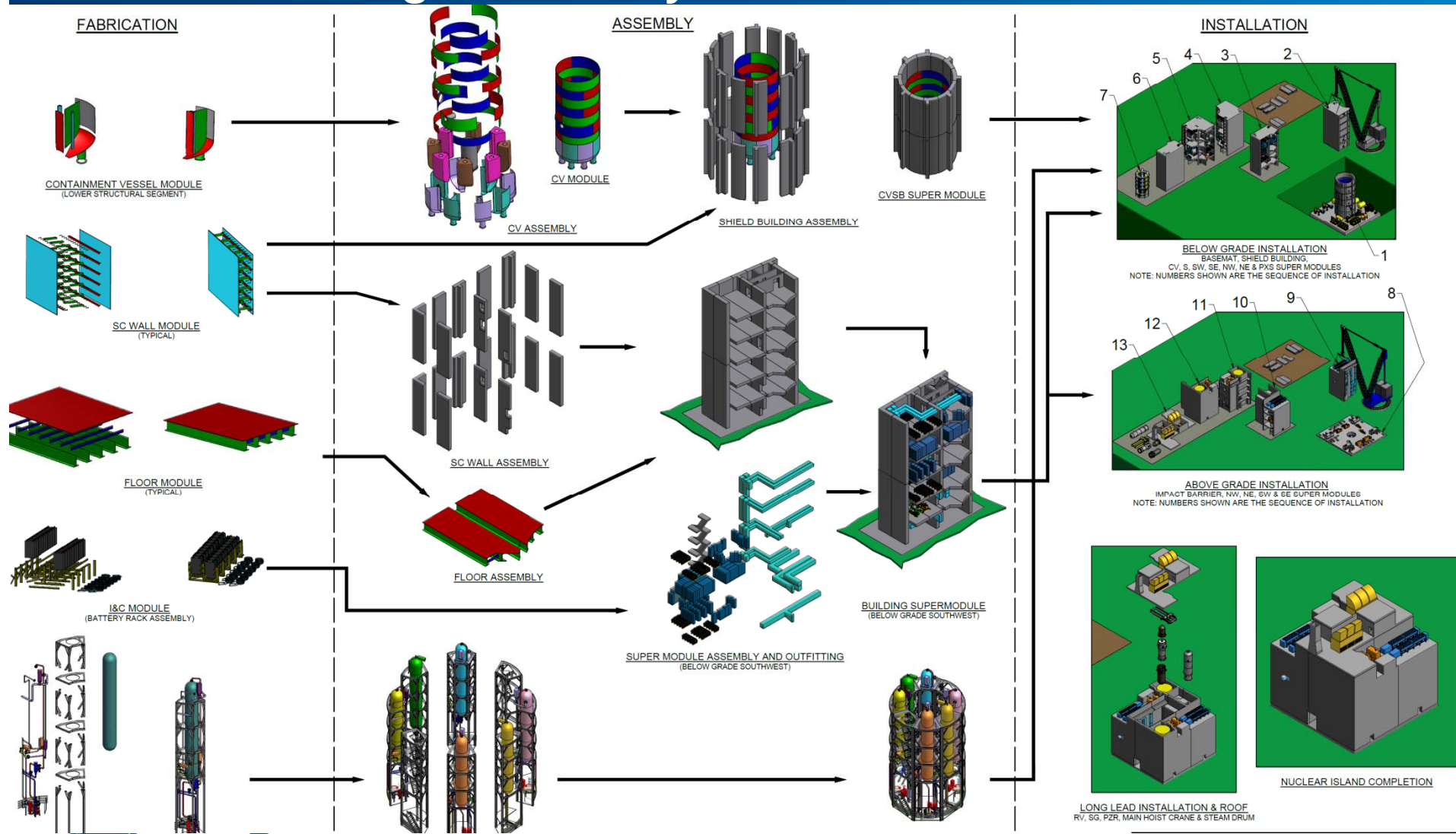


Westinghouse SMR Modularization Leverage Westinghouse and Industry Experience

- Steel composite panels (wall and floor modules)
- Mechanical modules (that is, equipment, tanks)
- Piping, electrical, HVAC modules
- Off-the-shelf pre-manufactured building modules



Westinghouse SMR Manufacturing Assembly Plan



Westinghouse SMR Modularization Nuclear Island Modularization

AP1000 Plant Composite Wall Module Structural Module



Westinghouse SMR Outfitted functional Super Module



Westinghouse SMR Taking AP1000 Plant Modules to the Next Tier



Westinghouse SMR Modularization Turbine Building Modularization

**AP1000 Plant
Turbine Building
Structural Module**



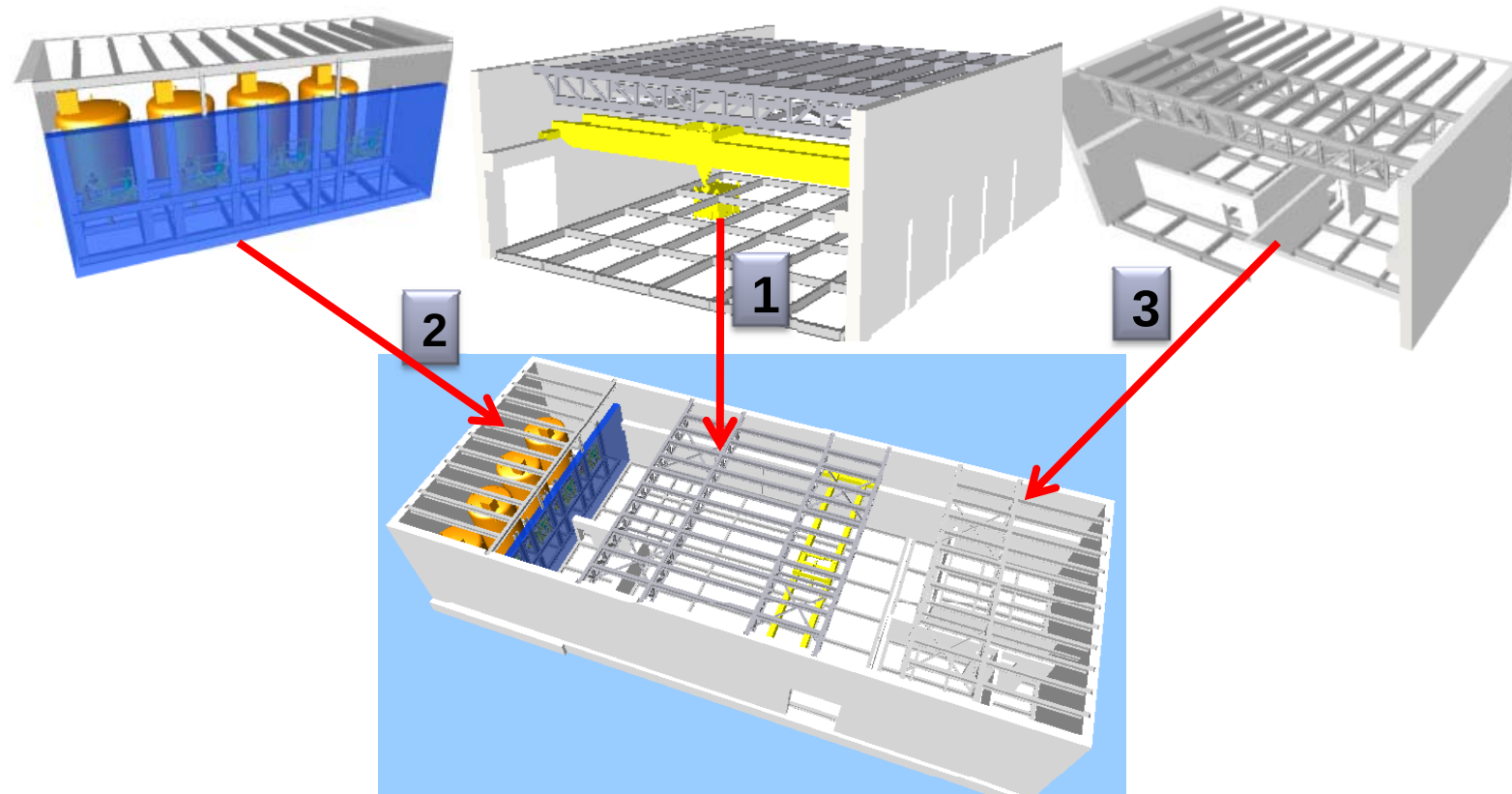
©2014 Sanmen Nuclear Power Company Ltd.
All rights reserved.

**Westinghouse SMR
Turbine Building outfitted
Super Equipment Module**



Westinghouse SMR Taking AP1000 Plant Modules to the Next Tier

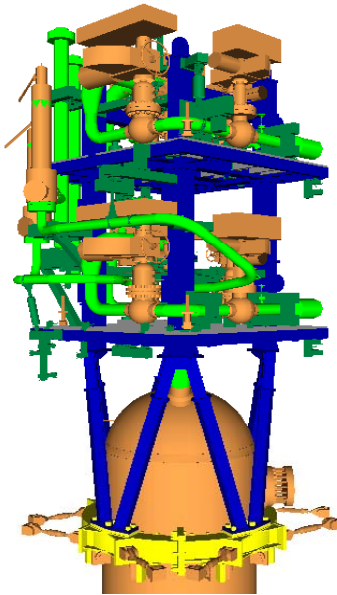
Westinghouse SMR Modularization Radwaste Building Modularization



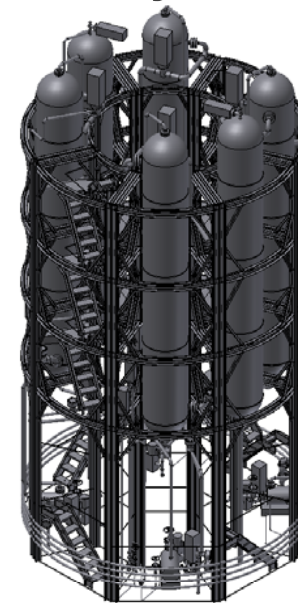
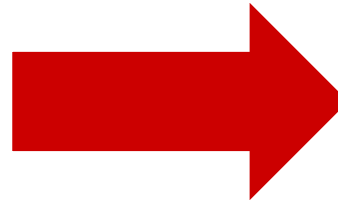
Modularize all Buildings with outfitted Modules

Westinghouse SMR Modularization Integrated Safety Modules

**AP1000 Plant Safety
Equipment Modules**



**Westinghouse SMR
Integrated Super Module
outfitted with most Safety
Systems**



Westinghouse SMR Taking AP1000 Plant Modules to the Next Tier

Summary – Module Advantages

- Safety
 - Improve construction site safety
- Cost
 - Major reduction in bulk materials
 - Reduce field work and onsite manpower
- Streamlined Construction Schedule
 - Predictable and shorter manufacturing
 - Pre-assembly and testing of major components
 - Parallel construction activities
- Quality
 - Work performed in optimal conditions/techniques
 - Continues improvement



